MT1011 Engineering Materials

Crystal Structures

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Crystal structures Part A

Learning outcomes

At the end of the section, you should be able to:

- Describe the crystalline structure
- Differentiate between a crystal and amorphous structures.
- Explain the concept of the unit cell.
- Explain lattice constants
- Describe the seven crystal systems using lattice parameters.
- Describe the 14 Bravais lattices.

Crystal structure

- Most engineering materials are crystalline
- Atoms arranged in a regular and repeating manner

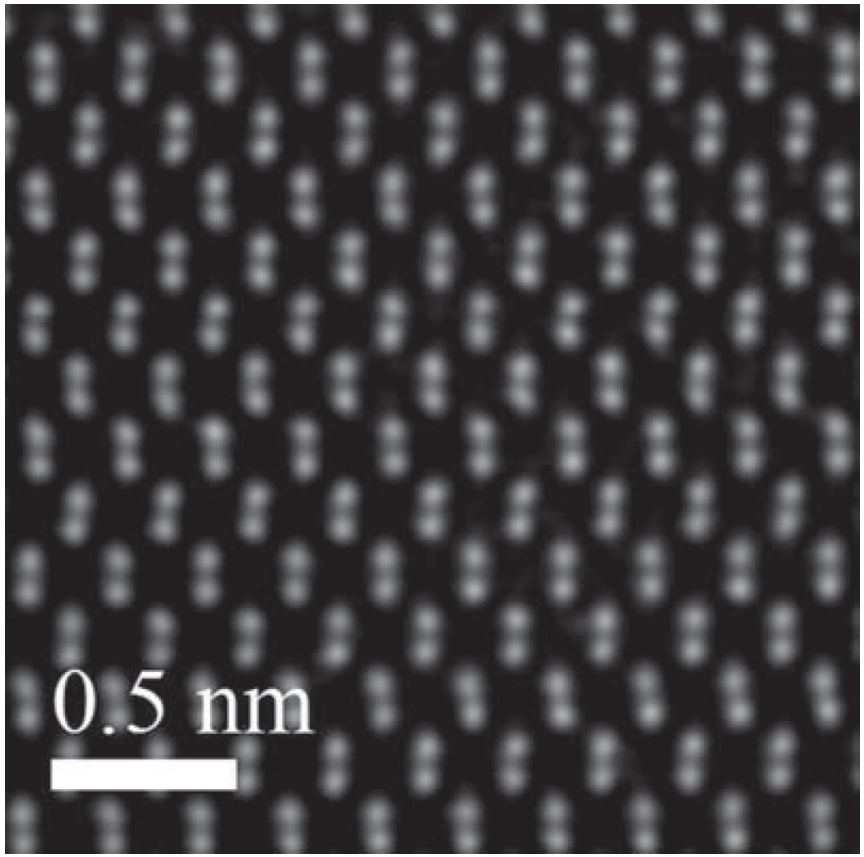


Figure: High-resolution transmission electron micrograph showing the repeating arrangement of atoms in a silicon crystal.
(Introduction to MATERIALS SCIENCE FOR ENGINEERS, James F. Shackelford, Micrograph credit: Klaus van Benthem and Andrew Thron, University of California, Davis.)

Crystal types

- There are 7 crystal-systems and the 14 crystal-lattices.
- Metals: Only four relatively simple types.
- Ceramic compounds: wide variety of crystalline structures, some simple some complex.
- Polymers: Crystalline polymers have small up to 50 volume % ordered regions. Glass: Non-crystalline (amorphous)

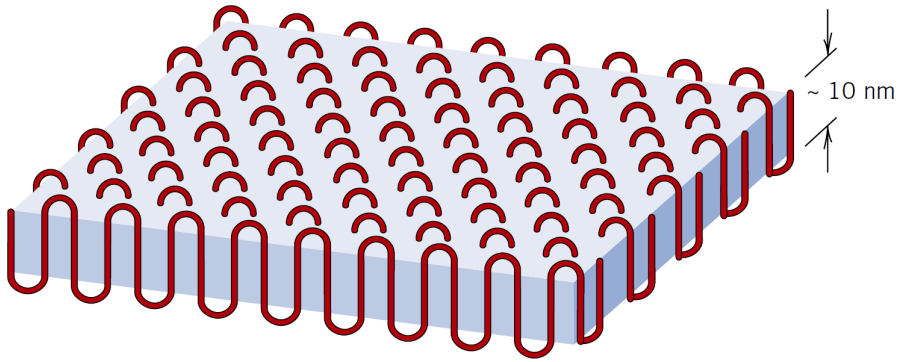
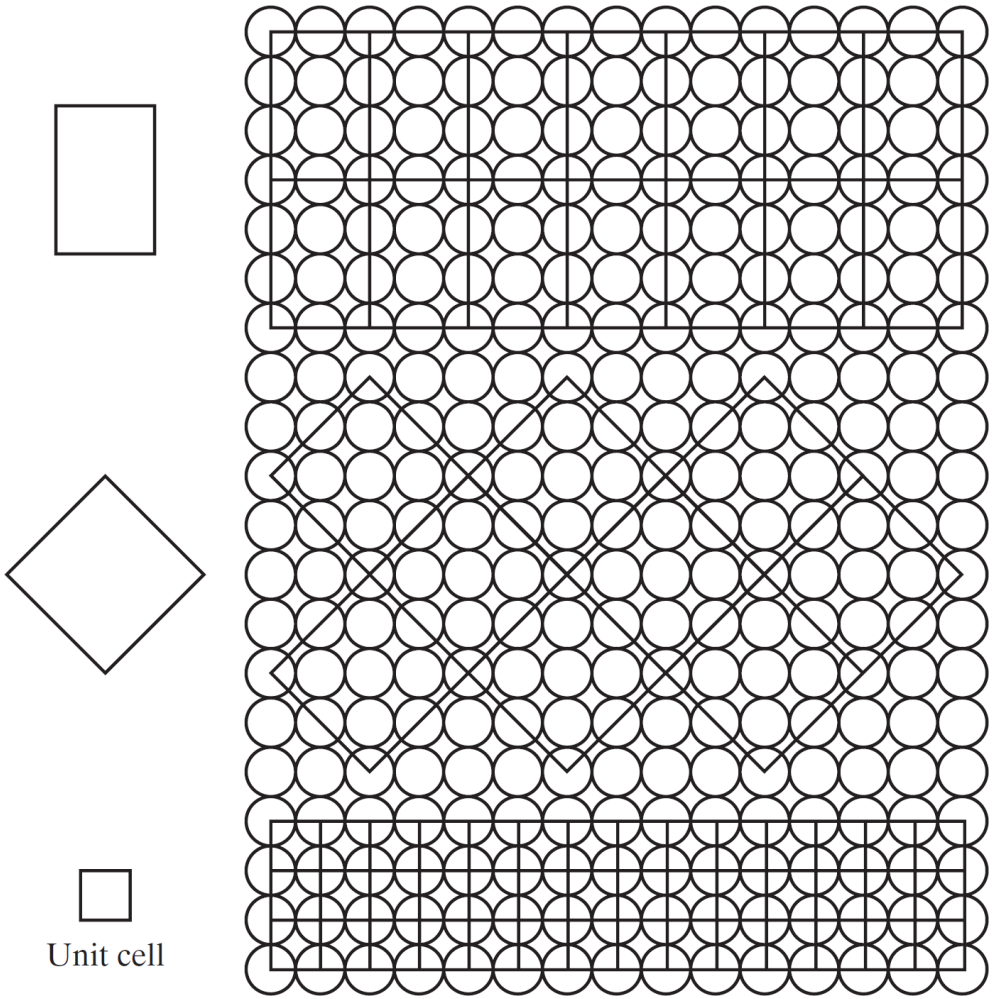


Figure: Chain-folded, plateshaped polymer crystallite.

The Unit cell

Among the various structural units that describe the crystalline structure, the simplest unit is the unit cell.



Unit cell

- **Lattice constants**, or **lattice parameters**: The length of unit-cell edges and the angles between crystallographic axes.
- The unit cell gives a full description of the whole structure.
- The entire crystal can be generated by **repeating** unit cells in the 3-D space.

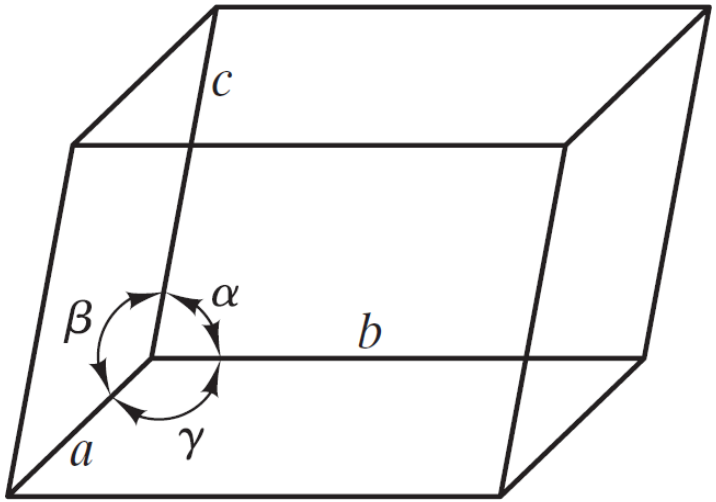
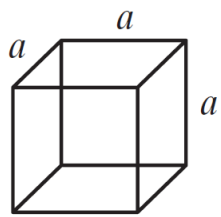
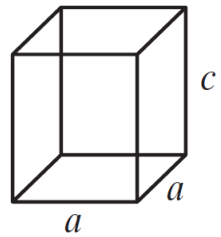
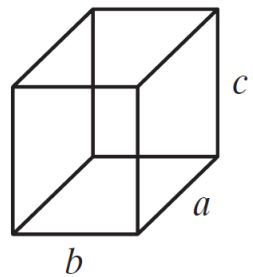


Figure: Unit cell geometry

The seven crystal systems

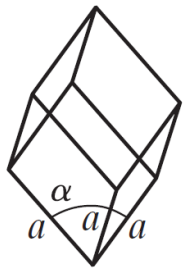
Only 7 unique unit-cell **shapes** that can be stacked together to fill three-dimensional space.

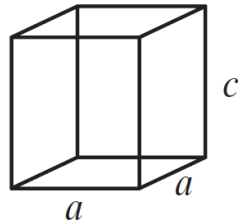
The Seven Crystal Systems

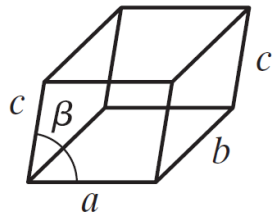
System	Axial lengths and angles[a]	Unit cell geometry
Cubic	$a = b = c,\ \alpha = \beta = \gamma = 90°$	
Tetragonal	$a = b \neq c,\ \alpha = \beta = \gamma = 90°$	
Orthorhombic	$a \neq b \neq c,\ \alpha = \beta = \gamma = 90°$	

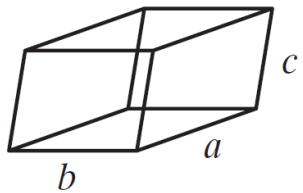
The seven crystal systems (continued)

Rhombohedral	$a = b = c,\ \alpha = \beta = \gamma \neq 90°$
Hexagonal	$a = b \neq c,\ \alpha = \beta = 90°,\ \gamma = 120°$
Monoclinic	$a \neq b \neq c,\ \alpha = \gamma = 90° \neq \beta$
Triclinic	$a \neq b \neq c,\ \alpha \neq \beta \neq \gamma \neq 90°$



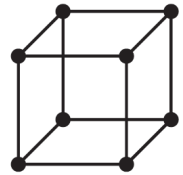


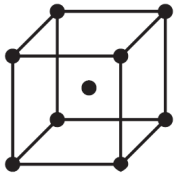


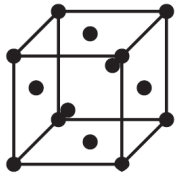


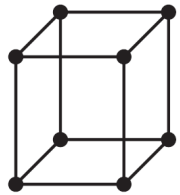
The 14 Bravais lattices

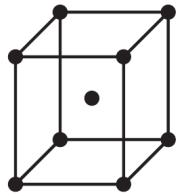
Atoms (lattice points) can be stacked together within a unit cell in a limited number of ways given by Bravais lattices.

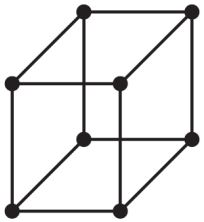

Simple cubic

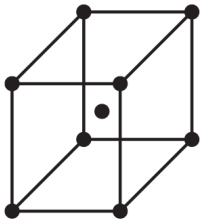

Body-centered cubic

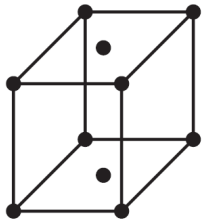

Face-centered cubic

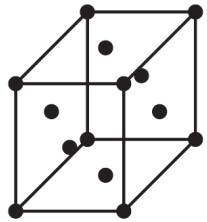

Simple tetragonal

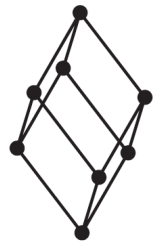

Body-centered tetragonal

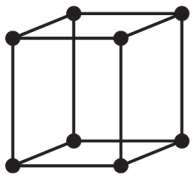

Simple orthorhombic

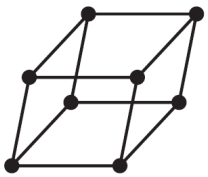

Body-centered orthorhombic

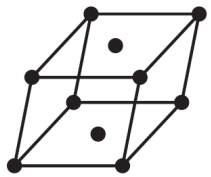

Base-centered orthorhombic

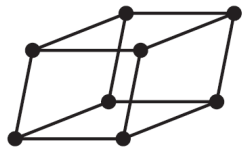

Face-centered orthorhombic


Rhombohedral


Hexagonal


Simple monoclinic


Base-centered monoclinic


Triclinic

Checkpoint

- What is a crystal?
- What materials form crystals?
- What is a unit cell?
- What are lattice constants?
- What is the difference between crystal systems and Bravais lattices?



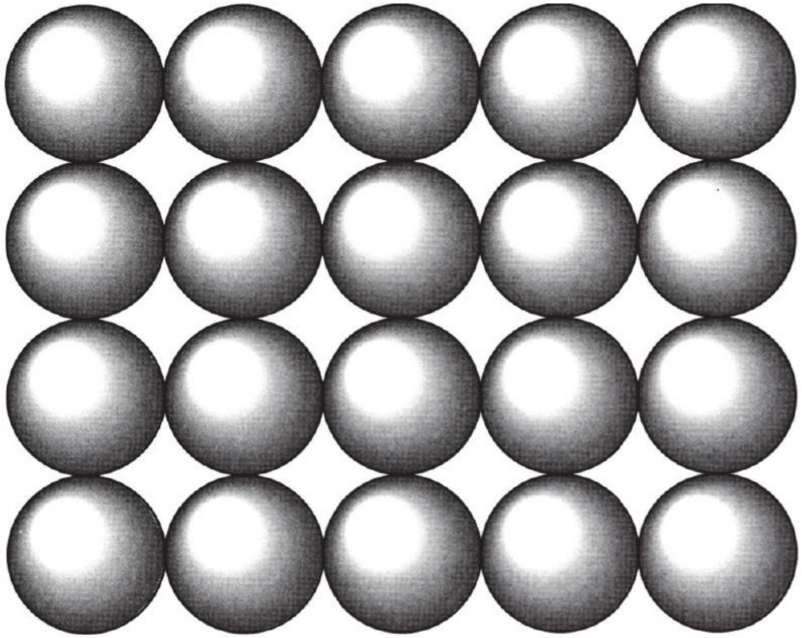
At the end of the section, for the simple cubic structure you should be able to:

- Describe the crystal structure
- Determine the number of atoms in each type of unit cell
- Explain the relationship between the atomic radius and lattice parameter
- Explain what is meant by the coordination number
- Calculate the atomic packing factor



Crystalline structure

Atoms as hard speres

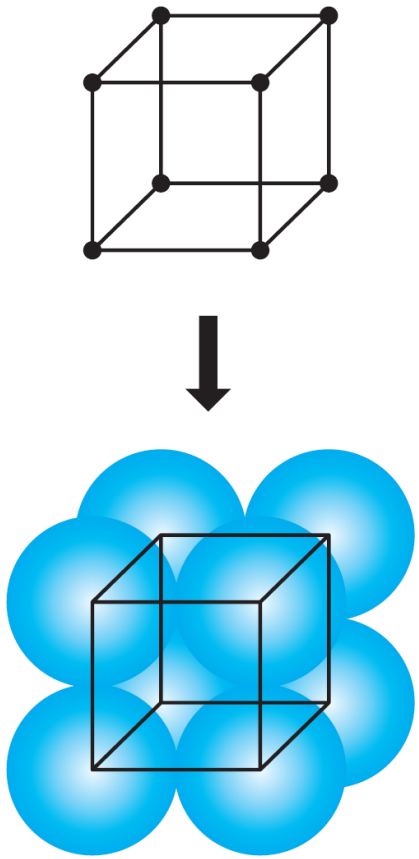


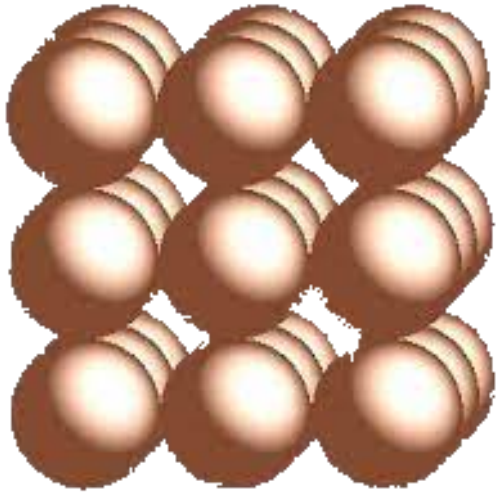
- Atoms can be thought of as hard spheres (hard sphere model)
- A layer of atoms can be arranged in the way shown that will reduce the overall volume of a crystal.
- It is possible to stack such layers on top of each other to form a simple cubic (SC) crystal structure.

Simple cubic structure

Unit cell

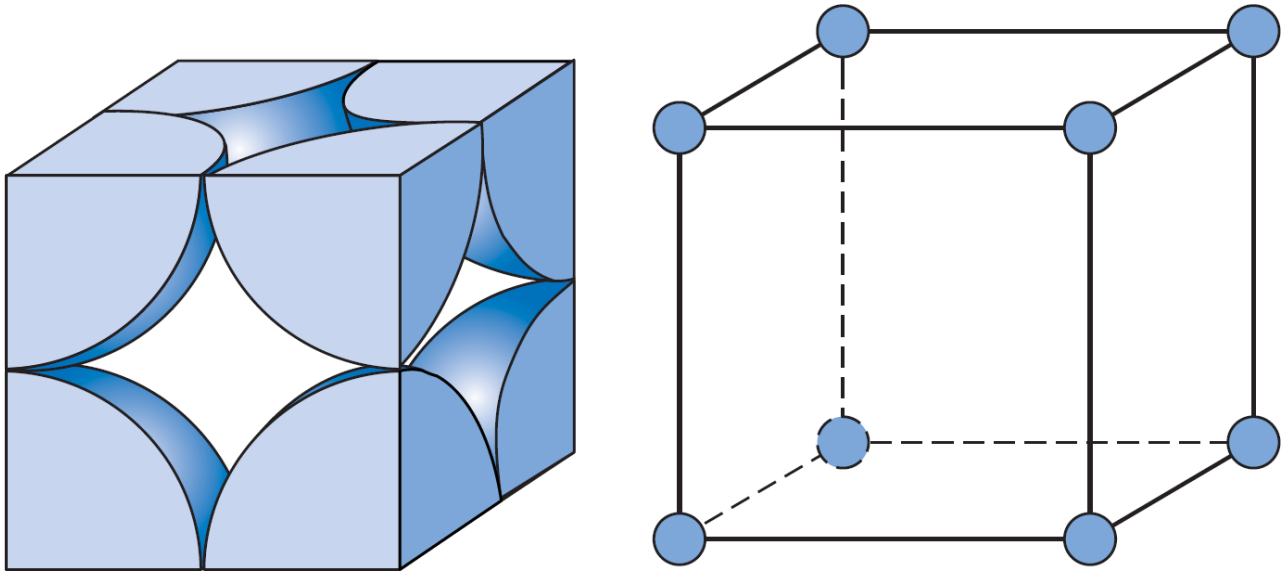
The simple cubic lattice has atoms at the corners of the unit cells





Simple cubic structure

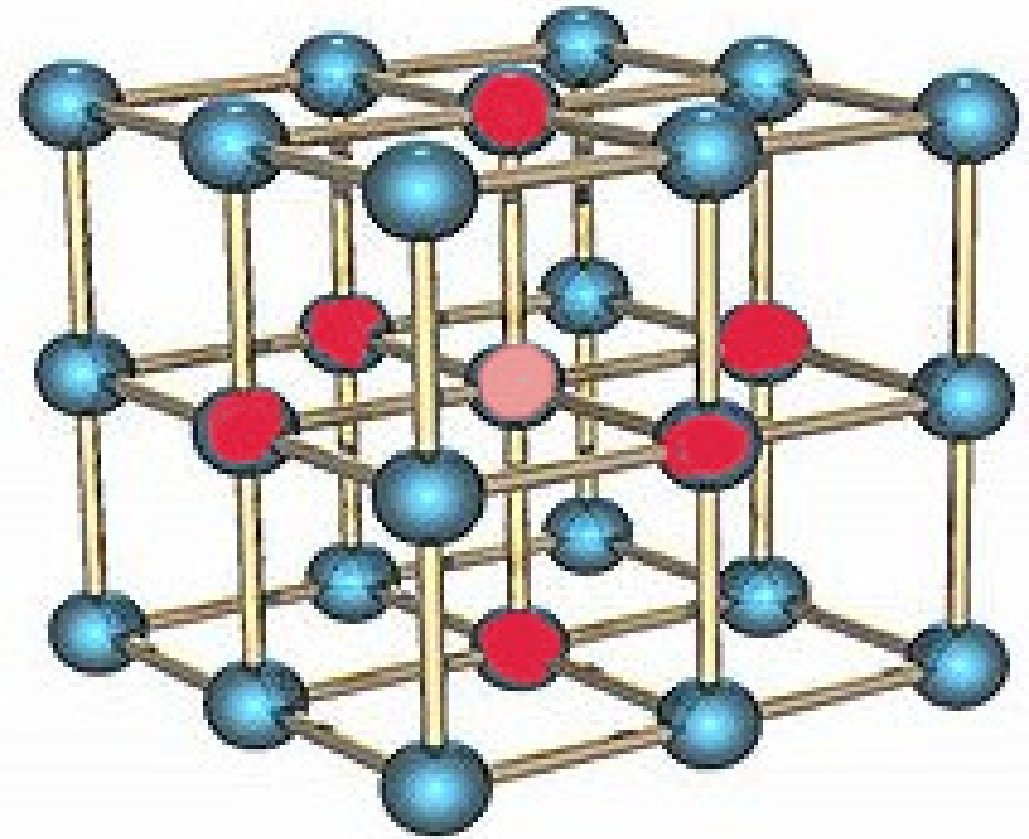
Atomic contribution to the unit cell



- Each corner atom contributes $\frac{1}{8}$ of an atom to the unit cell
- SC unit cell has 1 atom per unit cell $(8 \times \frac{1}{8})$.

Simple cubic structure

Coordination number



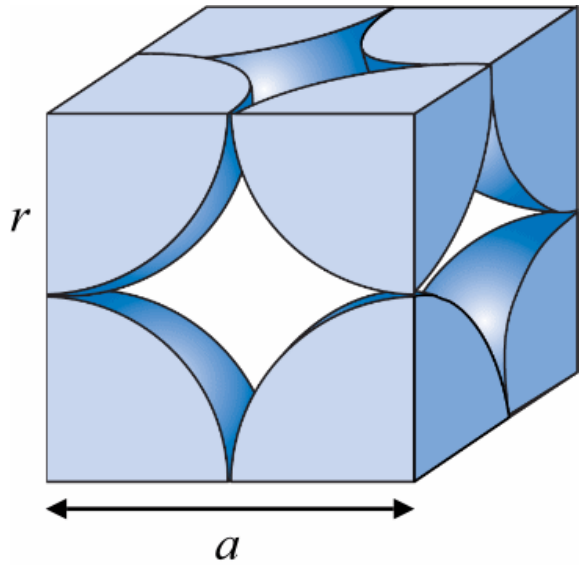
Each atom has 6 nearest neighbours (Coordination number = 6)

Simple cubic structure

Relatioship between a and r

Question

What is the relationship between the lattice parameter a and atomic radius r of the simple cubic structure?



Atoms touch along the edge of the cubic unit cell. Therefore,

$$a = 2r$$

Atomic packing factor (APF): Fraction of unit cell volume occupied by atoms

Question

What is the atomic packing factor of a SC crystal?

Because there is one atoms in the unit cell ($8 \times \frac{1}{8}$), the volume V_a occupied by atoms:

$$V_a = 1 \times \left(\frac{4}{3}\pi r^3\right) = \frac{4}{3}\pi r^3$$

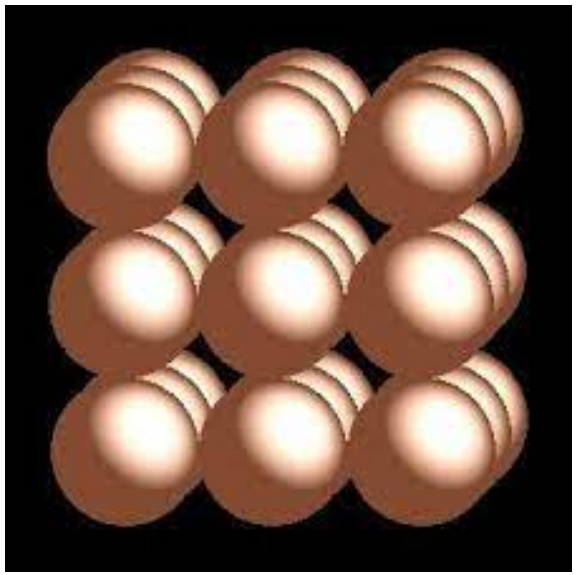
Volume of the unit cell V

$$V = a \times a \times a = a^3 = (2r)^3 = 8r^3$$

$$\text{APF} = \frac{V_a}{V} = \frac{4\pi r^3}{3 \times 8r^3} = \frac{\pi}{6} = 0.5236$$

Summary

- Simple cubic structure is one of the simplest but rare.
- Polonium (Po) is an example.
- Atomic packing is about 52% (Only half the volume is occupied by atoms)
- Each atom in the structure has six neighbours.



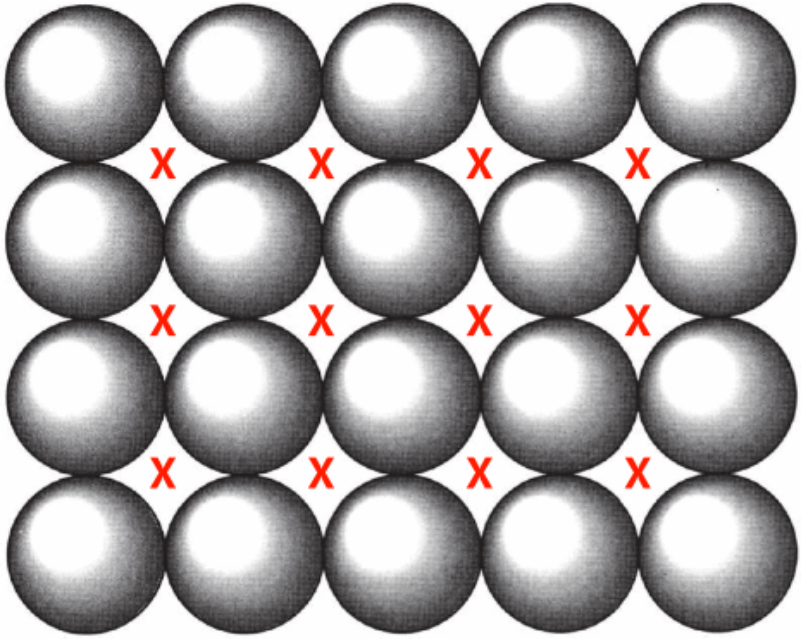
At the end of the section, for the body centred cubic structure you should be able to:

- Describe the crystal structure
- Determine the number of atoms in each type of unit cell
- Explain the relationship between the atomic radius and lattice parameter
- Explain what is meant by the coordination number
- Calculate the atomic packing factor



Body centre cubic structure

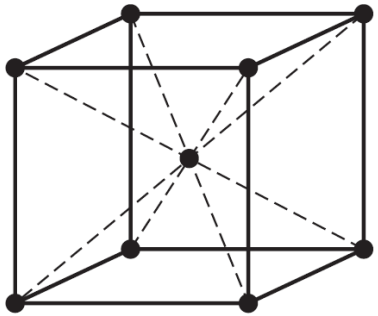
Atomic layers

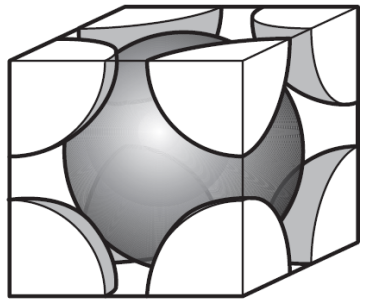


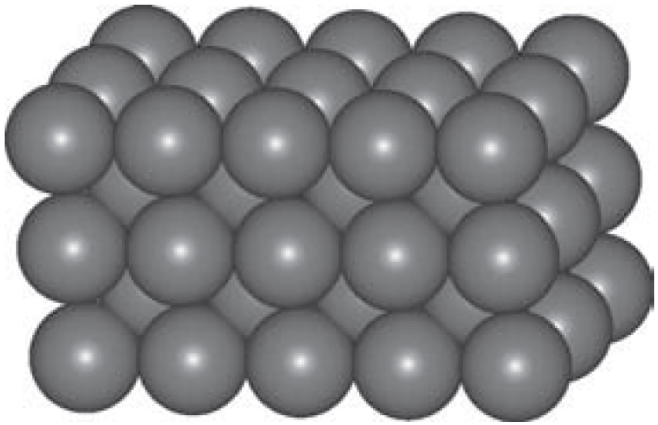
- If the second layer of atoms goes into locations marked X better packing can be achieved compared to the SC structure.
- The third layer can go directly above the first layer.
- When the layers repeat this way, it will give the body center cubic (BCC) crystal structure.

Body centre cubic structure

Unit cell



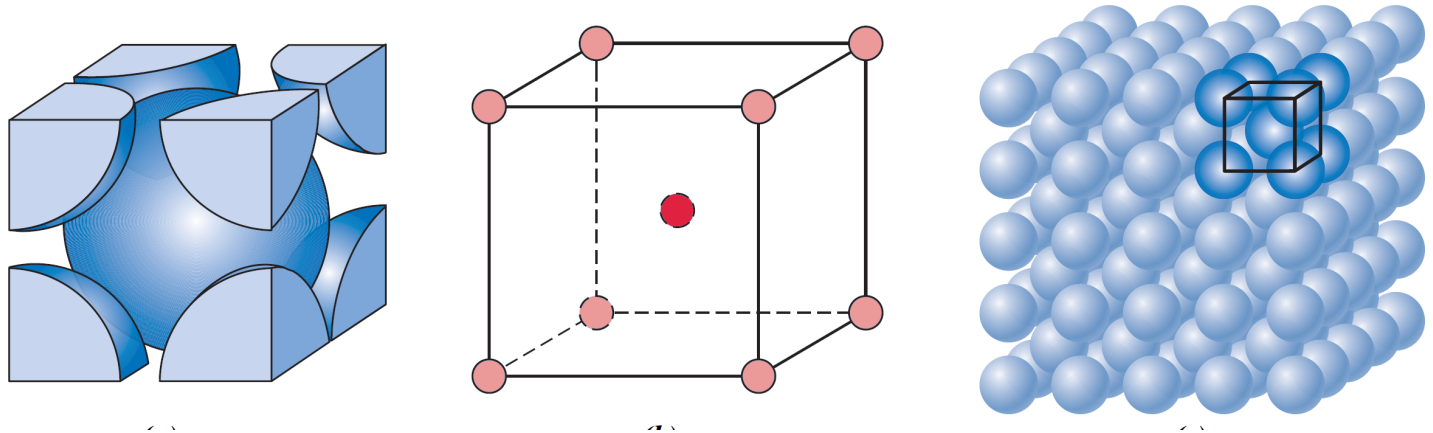




- Atoms at 8 corners and one at the cube center (body center).
- Two atoms per unit cell $(8 \times \frac{1}{8}) + 1$.

Question

What is the coordination number for the BCC crystal?



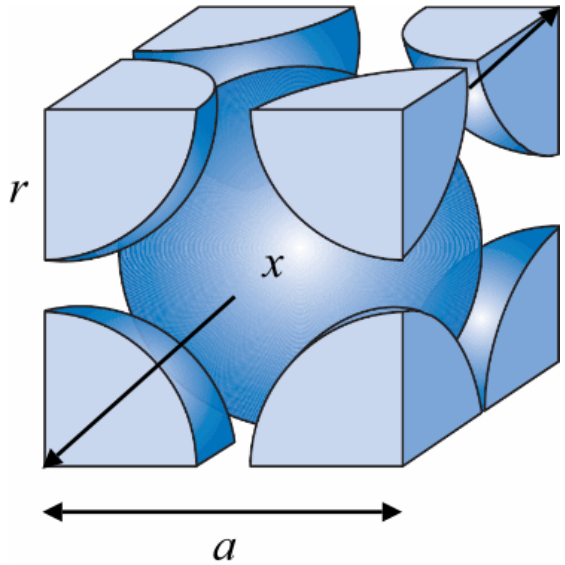
BCC atom has eight nearest neighbours (pink)
(Coordination number = 8)

Body centre cubic structure (BCC)

Relatioship between a and r

Question

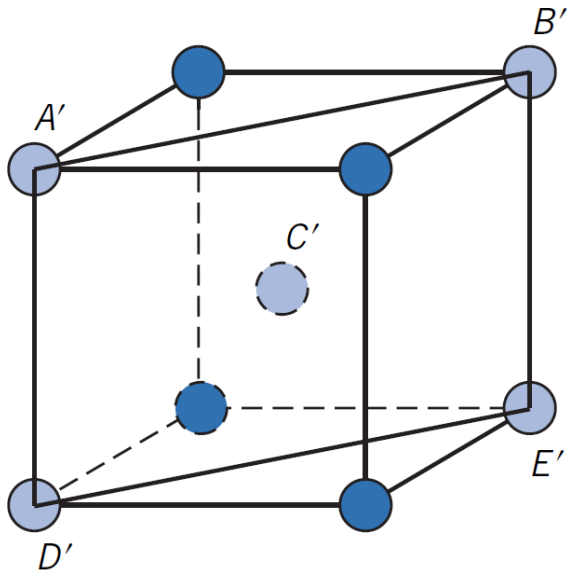
What is the relationship between the lattice parameter a and atomic radius r of the BCC structure?

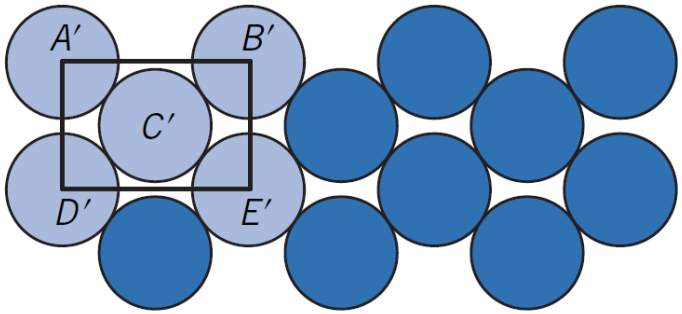


Atoms touch along the body diagonal of the cubic unit cell.

Body centre cubic structure (BCC)

Relatioship between a and r





Applying Pythagoras' theorem twice

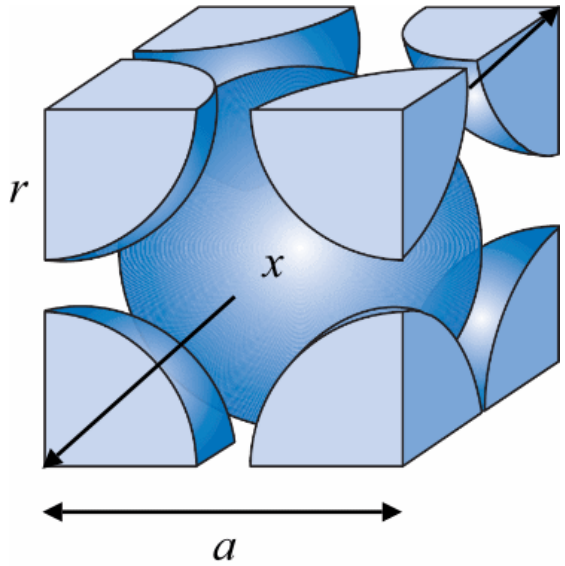
$$A'B' = \sqrt{a^2 + a^2} = \sqrt{2}a$$

$$D'B' = \sqrt{(A'B')^2 + (A'D')^2} = \sqrt{2a^2 + a^2} = \sqrt{3}a$$

Length of body diagonal $= \sqrt{3}a$

Body centre cubic structure (BCC)

Relatioship between a and r



$$\sqrt{3}a = x = (r + 2r + r) = 4r$$

$$a = \frac{4}{\sqrt{3}}r$$

Question

What is the atomic packing factor of a BCC crystal?

Because there are two atoms in the unit cell ($8 \times \frac{1}{8} + 1$), the volume V_a occupied by atoms:

$$V_a = 2 \times \left(\frac{4}{3}\pi r^3\right) = \frac{8}{3}\pi r^3$$

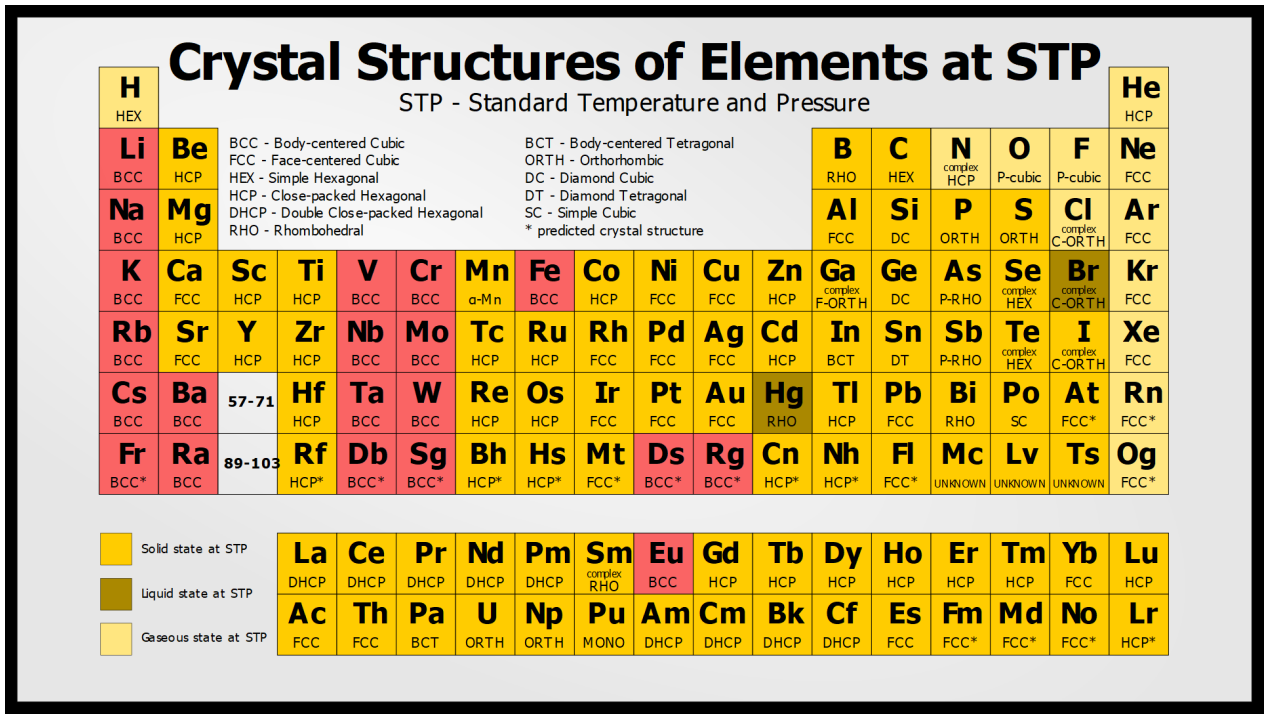
Volume of the unit cell V

$$V = a^3 = (\frac{4}{\sqrt{3}}r)^3 = 8r^3$$

$$\text{APF} = \frac{V_a}{V} = \frac{\sqrt{3}\pi}{8} = 0.6802$$

Summary

- BCC structure is very common.
- Examples: Fe (below $910\,^{\circ}\mathrm{C}$), Cr, Li, Na, K, Nb, Mo, Ta, W.
- Atomic packing is about 68% much more packing than in SC.
- Each atom in the structure has eight neighbours.



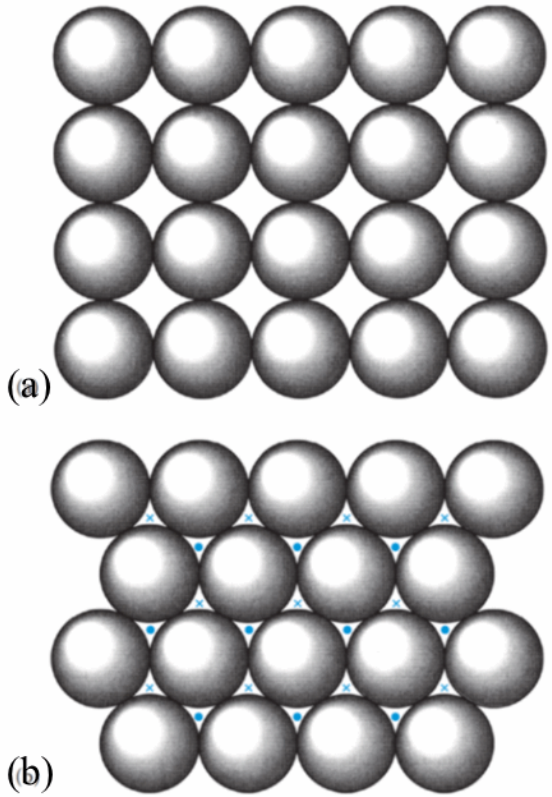
At the end of the section, you should be able to:

- Illustrate the unit cells of face centre cubic and hexagonal close packed crystal structures
- Describe the FCC unit cell, coordination number and atomic packing factor.
- Explain the HCP unit cell.



Packing of atomic layers

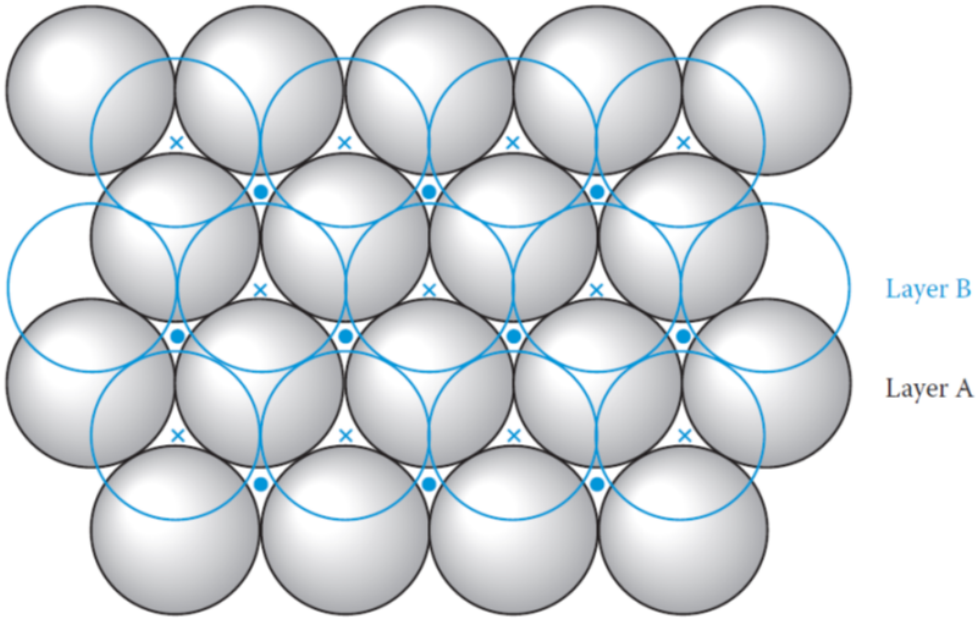
Two possibilities for close packing



- Type (a) is the basis for SC and BCC structures.
- Type (b) is the basis of FCC (face centre cubic) and HCP (hexagonal close packed) structures.

Packing of atomic layers

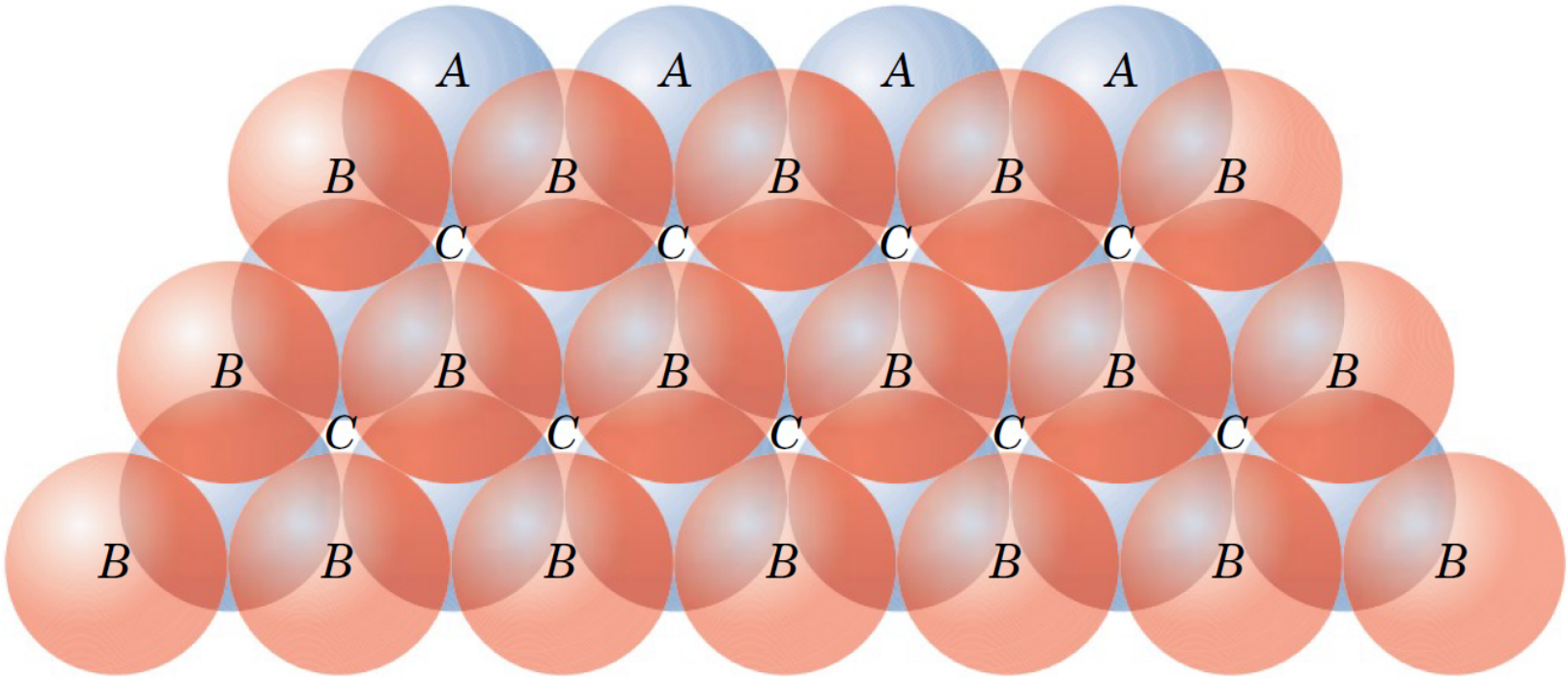
Close packed structures FCC and HCP



- The third layer will decide whether FCC or BCC structures are possible.
- ABABAB ... type layering will five the HCP structure.
- ABCABCABC ..... layering wil give the FCC structure.

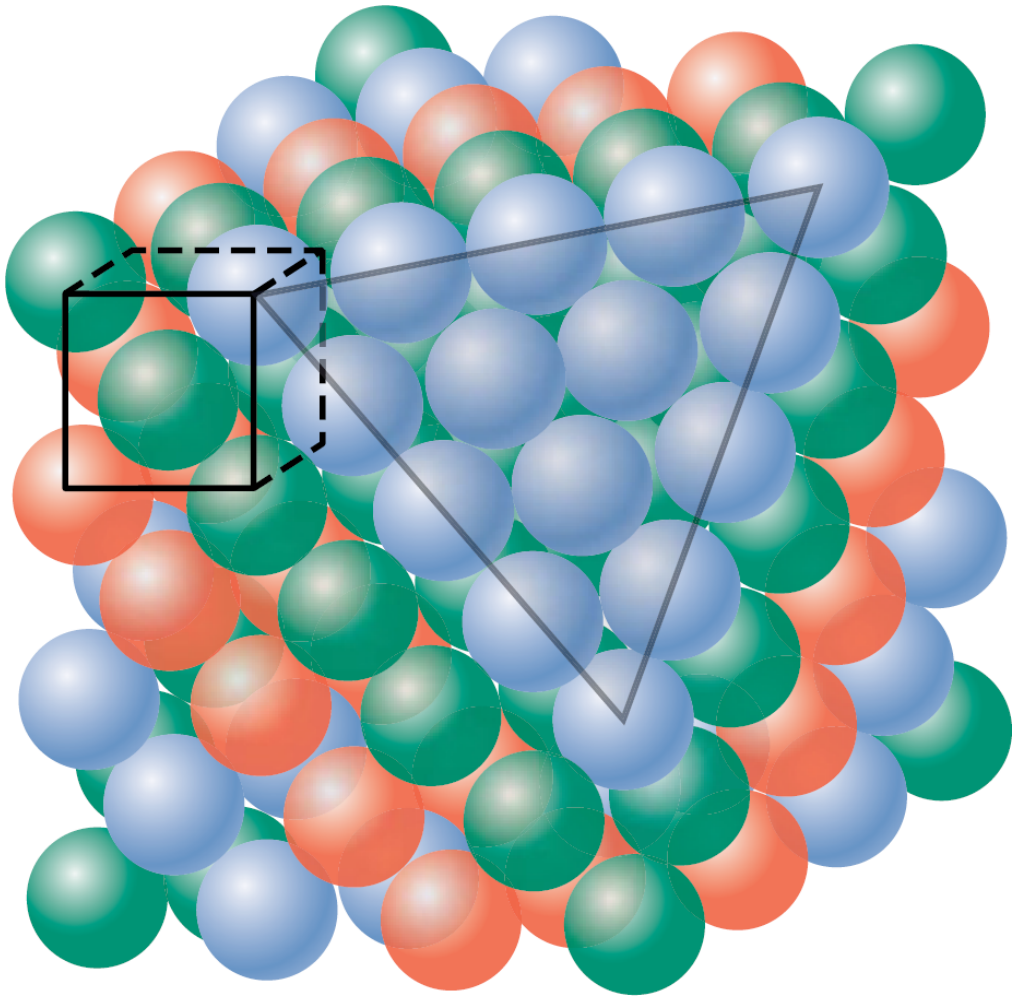
Face centred cubic structure

Atomic packing



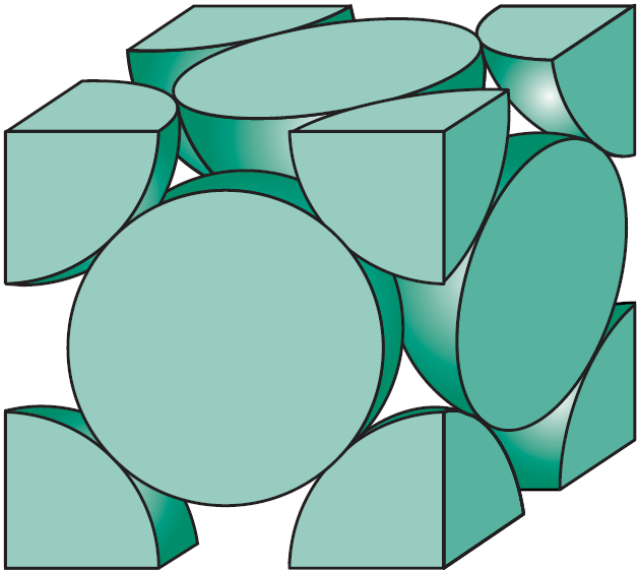
Face centred cubic structure

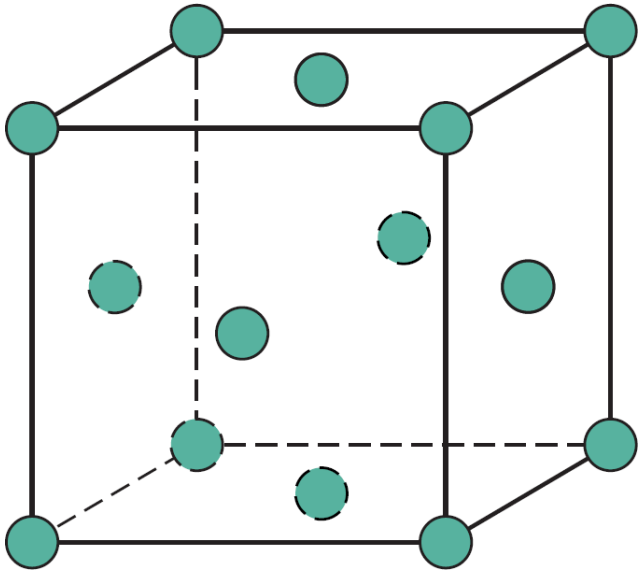
Atomic packing



Face centre cubic structure (FCC)

Atoms per unit cell

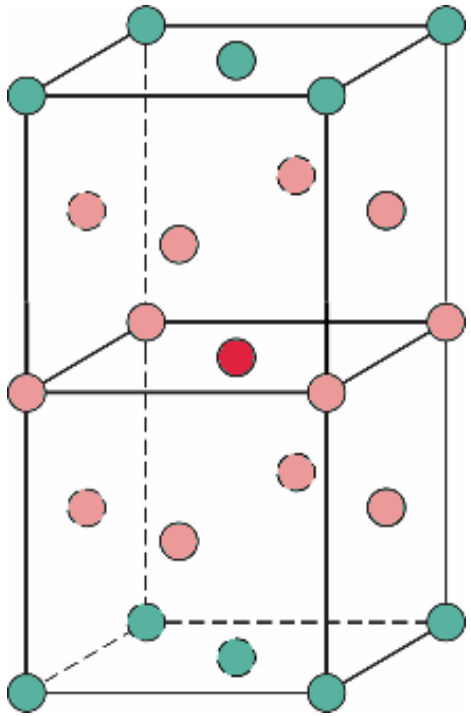




- Atoms at **8 corners** and one at the centre of **6 faces**.
- Each corner atom contributes $\frac{1}{8}$ of the atom to the unit cell.
- Each face center atom contributes $\frac{1}{2}$ an atom to the unit cell.
- Therefore, there are **4 atoms** in the unit cell.
  $(8 \times \frac{1}{8}) + (6 \times \frac{1}{2}) = 4$.

Face center cubic structure (FCC)

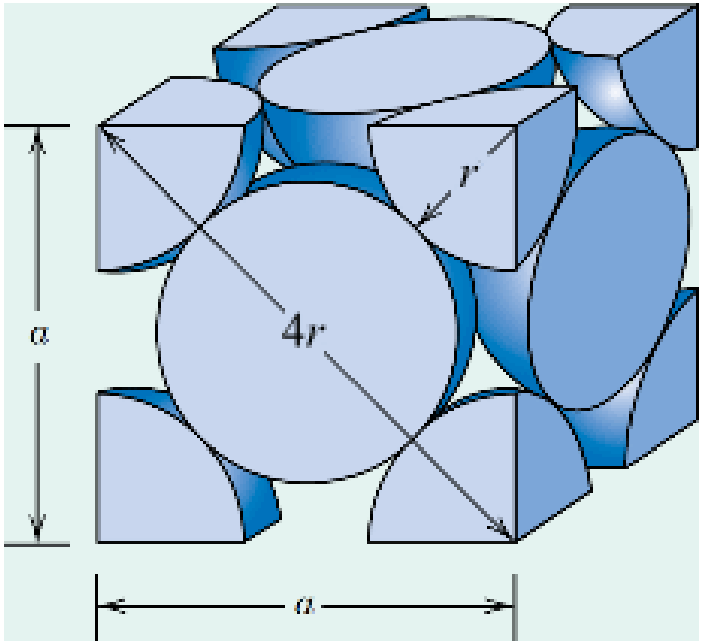
Coordination number



- Consider 2 unit cells as shown.
- Each atom (red) in the FCC structure has twelve nearest neighbours (pink)
- Coordination number = 12

Face centre cubic structure (FCC)

Relatioship between a and r



- Atoms touch along the body diagonal of the cubic unit cell.
- Applying Pythagoras' theorem to the right angled triangle formed by the two edges and the face diagonal of the unit cell:

$$a^2 + a^2 = (4r)^2$$

$$2a^2 = 16r^2$$

$$a = 2\sqrt{2}r$$

Face centre cubic structure (FCC)

Atomic packing factor

- The number of atoms in the unit cell $= 4$
- The volume V_a occupied by atoms

$$V_a = 4 \times \left(\frac{4}{3}\pi r^3\right) = \frac{16}{3}\pi r^3$$

- The volume of the unit cell V

$$V = a^3 = (2\sqrt{2}r)^3 = 16\sqrt{2}r^3$$

- Atomic packing factor is therefore

$$\text{APF} = \frac{V_a}{V} = \frac{\pi}{3\sqrt{2}} = 0.7405$$

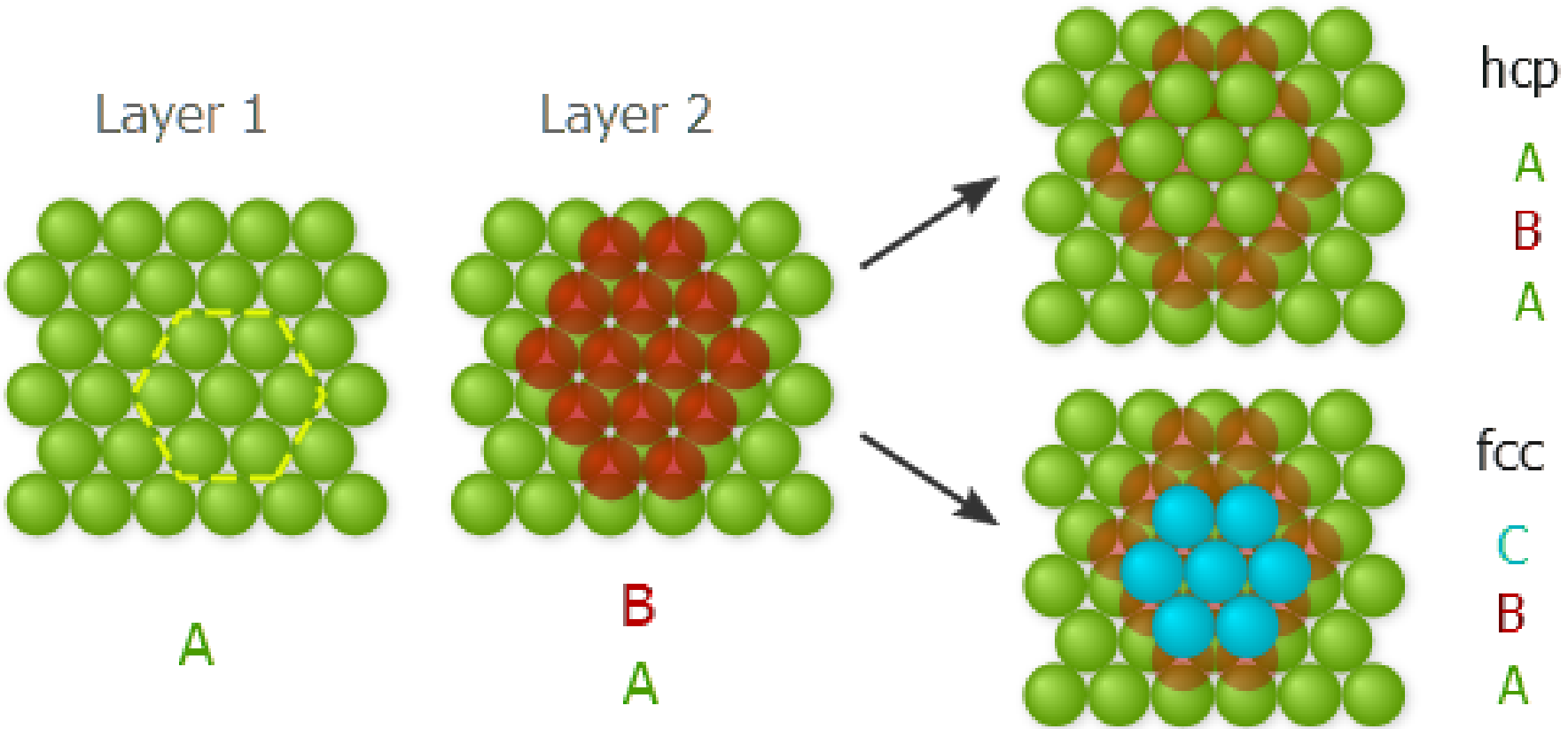


Parameter	SC	BCC	FCC
Atoms per cell	1	2	4
Coordination number	6	8	12
a/r	2	$4/\sqrt{3}$	$2\sqrt{2}$
Unit cell side	$2r$		
Face diagonal			$4r$
Body diagonal		$4r$	
Cell volume	$8r^3$		
APF	0.5236	0.6802	0.7405



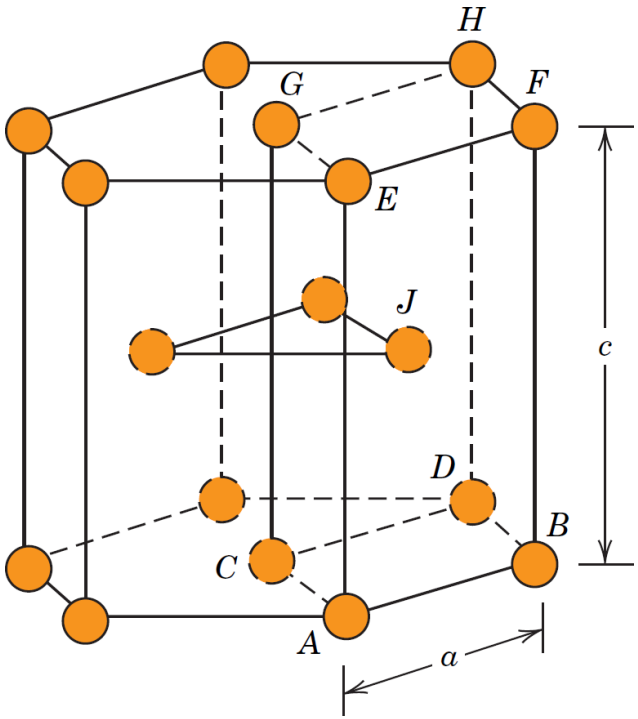
Hexagonal close packed structure (HCP)

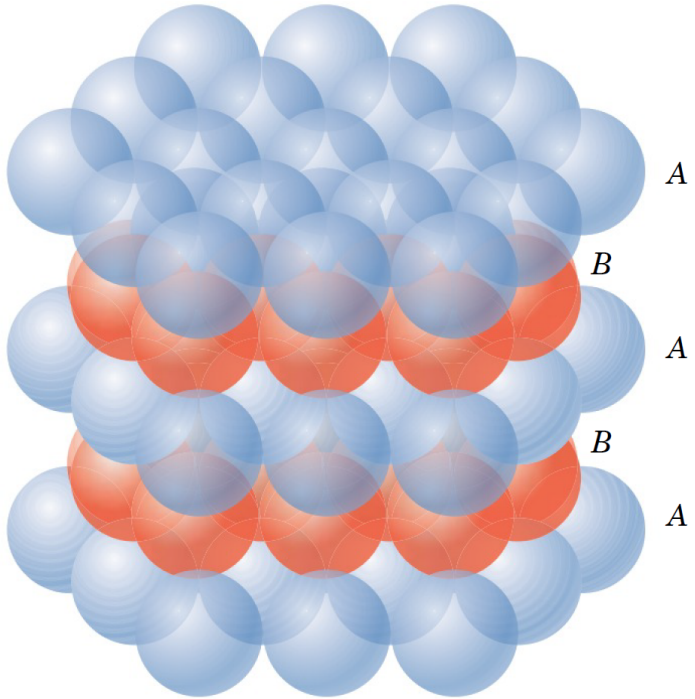
Atomic packing



Hexagonal close packed structure (HCP)

- Top and bottom faces have six atoms (regular hexagons) which surround a center atom.
- The middle plane has three more atoms.
- Midplane atoms have nearest neighbours in both top and bottom planes.



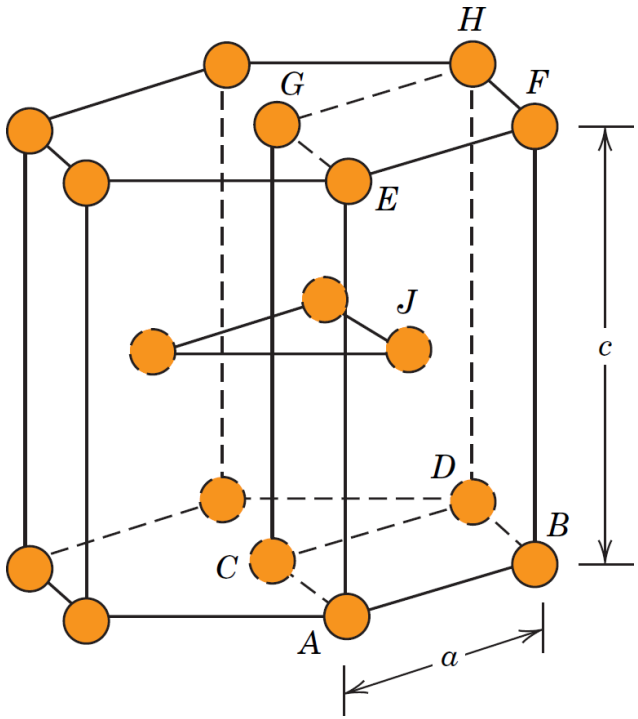


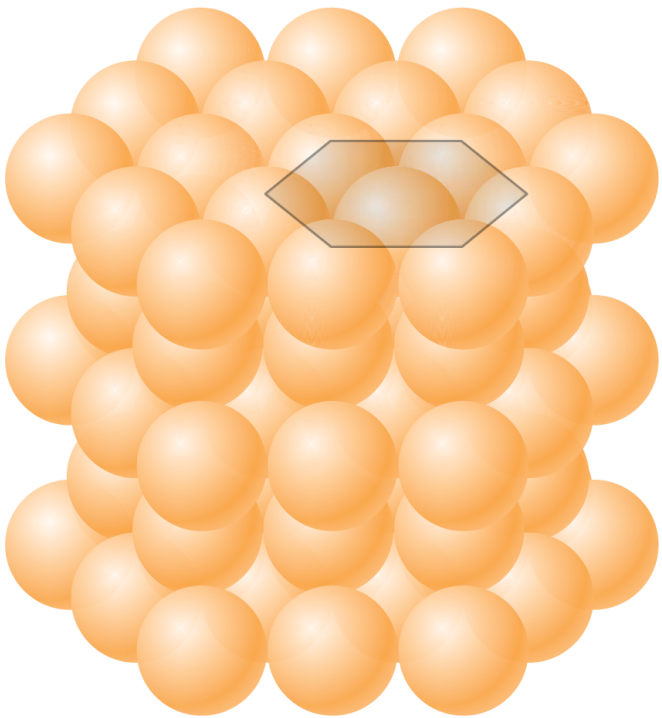
Hexagonal close packed structure (HCP)

Atoms per unit cell

For the unit cell

- Each corner atom (e.g. A) gives $\frac{1}{6}$ of an atom $(\frac{1}{6} \times 12 = 2)$
- Each centre atom (e.g. C) gives $\frac{1}{2}$ of an atom $(\frac{1}{2} \times 2 = 1)$
- Every mid-plane atom (e.g. J) gives a full atom $(1 \times 3 = 3)$
- Total atoms per unit cell $= 2 + 1 + 3 = 6$





- Points in the lattice (or unit cell) such as atom positions can be located using the **right-handed** coordinate system.
- Distances are measured in units of **lattice parameters** in each of the x, y, and z coordinates.
- The coordinates are written as the three distances, with **commas** separating the numbers.

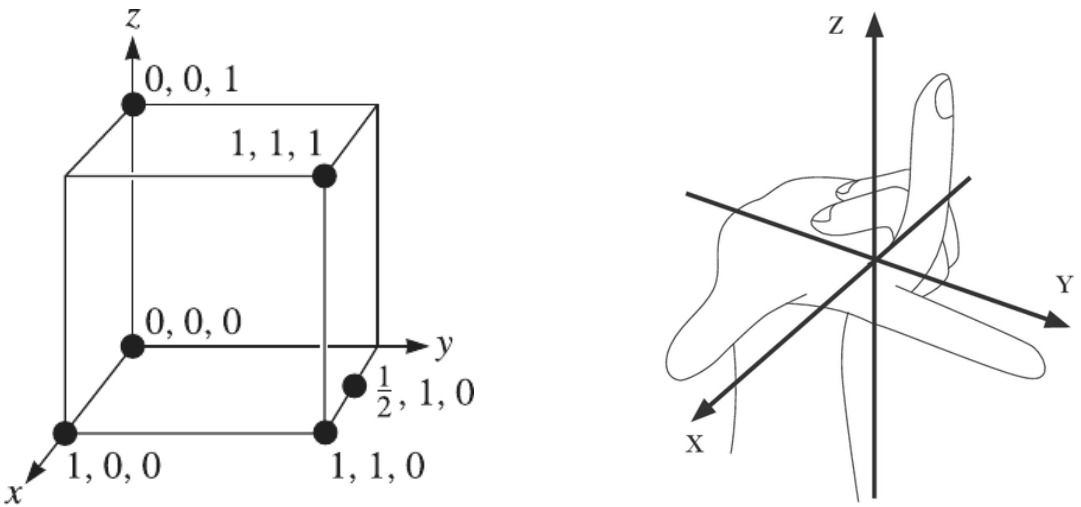
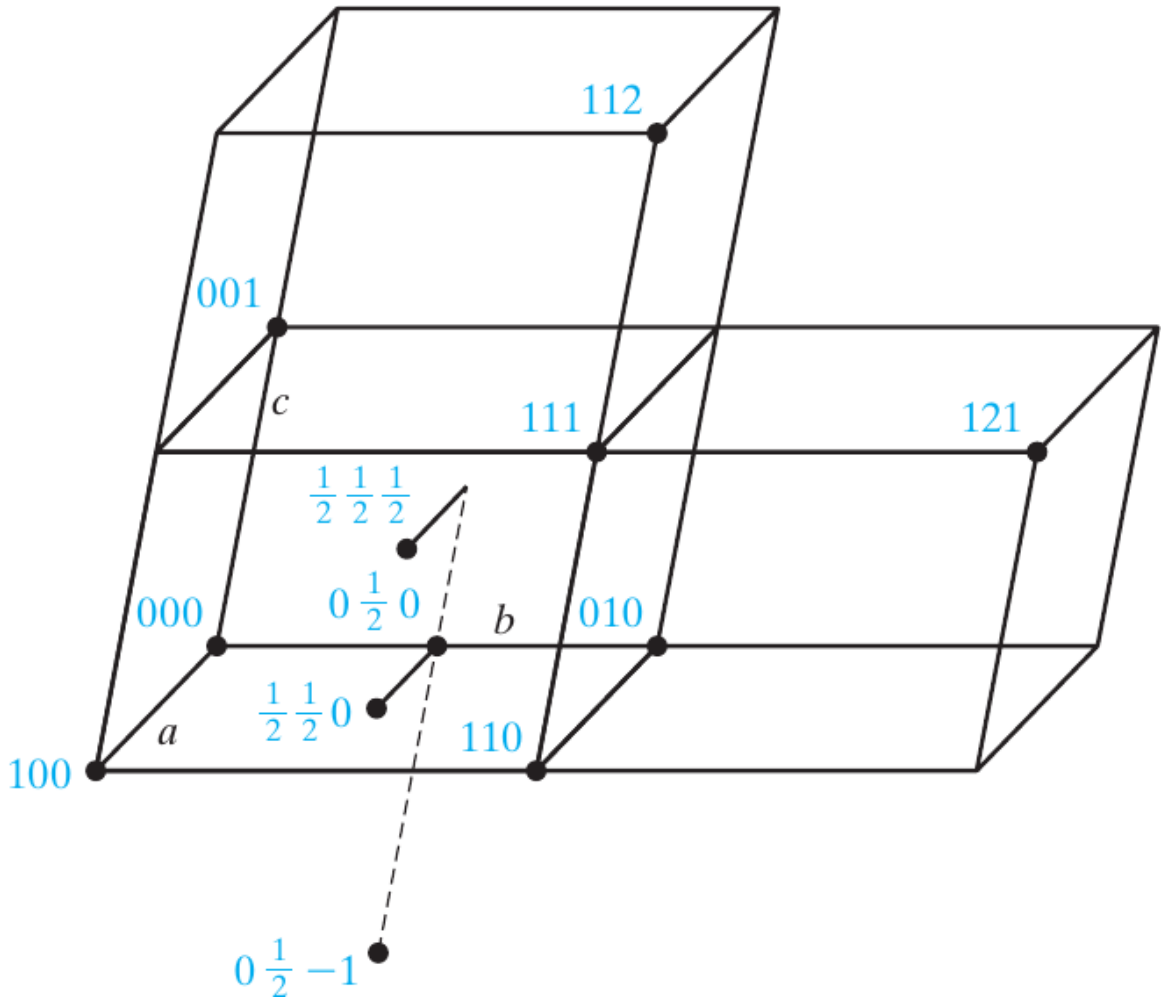


Figure: Coordinates of selected points in the unit cell using the right-handed coordinate system. The number refers to the distance from the origin in terms of lattice parameters.

Lattice positions are described as fractions or multiples of unit-cell dimensions.

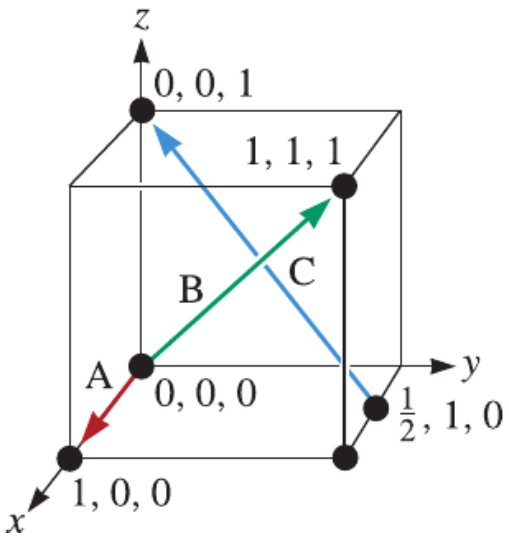


The indices for a direction is determined using a four-step procedure:

1. Using a right-handed coordinate system, determine the coordinates of two points that lie on the direction.
2. Subtract the coordinates of the **tail** point from the coordinates of the **head** point to determine the number of lattice parameters travelled in the direction of each axis.
3. Clear fractions and reduce the result to the nearest integer (so that 1.25 would become 1).
4. Enclose the numbers in **square** brackets [ ]. If a negative sign is produced, represent the negative sign with a **bar** over the number. (for example $[1\,\overline{2}\,0]$ would correspond to 1 in the x-direction, -2 in the y-direction, and 0 in the z-direction).



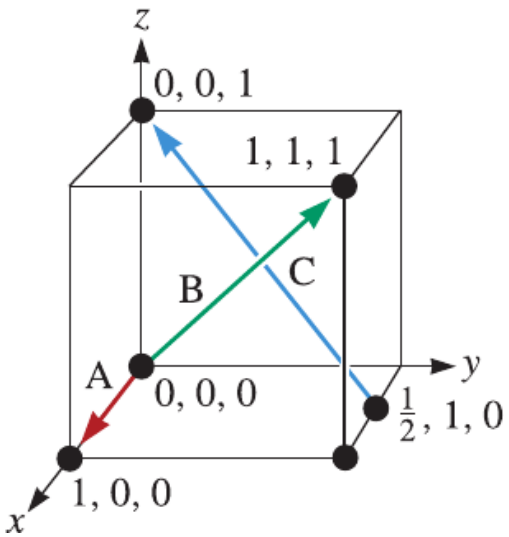
Determine the indices of directions A, B, and C.



Solution: Direction A

1. Two points are $1, 0, 0$ (head), and $0, 0, 0$ (tail)
2. (head) - (tail) $= 1, 0, 0 - 0, 0, 0 = 1, 0, 0$
3. No fractions to clear or integers to reduce
4. $[1\,0\,0]$

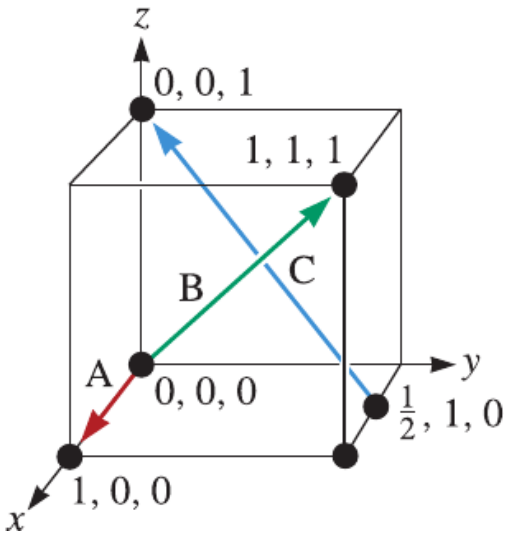
Determine the indices of directions A, B, and C.



Solution: Direction B

1. Two points are $1, 1, 1$ (head), and $0, 0, 0$ (tail)
2. (head) - (tail) $= 1, 1, 1 - 0, 0, 0 = 1, 1, 1$
3. No fractions to clear or integers to reduce
4. $[1\,1\,1]$

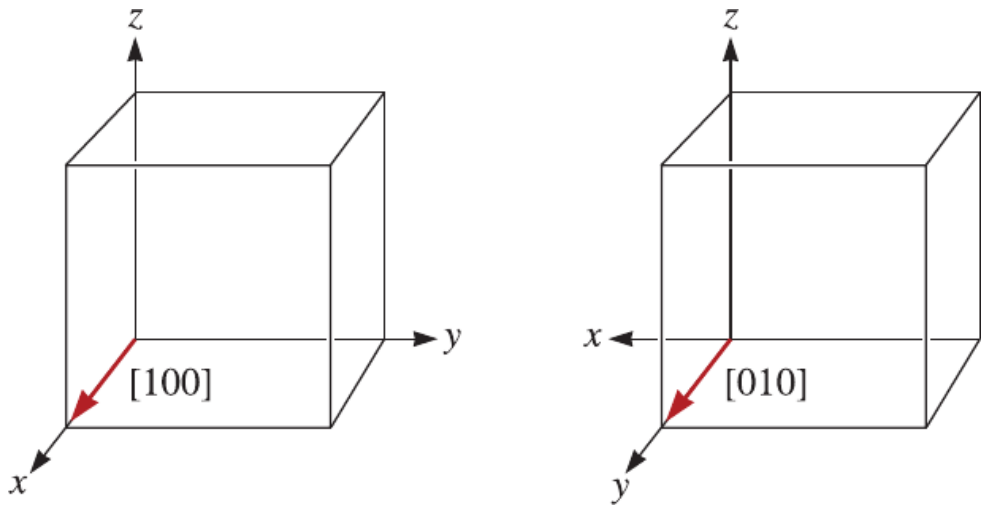
Determine the indices of directions A, B, and C.



Solution: Direction C

1. Two points are $0, 0, 1$ (head), and $\frac{1}{2}, 1, 0$ (tail)
2. (head) - (tail) $= 0, 0, 1 - \frac{1}{2}, 1, 0 = -\frac{1}{2}, -1, 1$
3. $2 \times (-\frac{1}{2}, -1, 1) = -1, -2, 2$
4. $[\bar{1}\,\bar{2}\,2]$

- Because directions are vectors, a direction and its negative are not identical. For example $[1\,0\,0]$ is not equal to $[\bar{1}\,0\,0]$. They represent the same line, but opposite directions.
- A direction and its multiple are identical. For example $[1\,0\,0]$ is the same direction as $[2\,0\,0]$.
- Certain groups of directions are equivalent. Their particular indices are due to the way we construct the coordinates. For example, in a cubic system, a $[1\,0\,0]$ direction is a $[0\,1\,0]$ direction if we redefine the coordinate system.



- Groups of equivalent directions form a family.
- The special brackets $\langle\ \rangle$ are used to indicate this collection of directions.
- All of the directions of the form $\langle 1\,1\,0 \rangle$ are listed below.
- The material have the same properties in each of these twelve directions of the form $\langle 1\,1\,0 \rangle$

Directions of the form $\langle 110 \rangle$ in cubic systems

$$\langle 110 \rangle = \begin{cases} [110]\,[\bar{1}\,\bar{1}0] \\ [101]\,[\bar{1}0\bar{1}] \\ [011]\,[0\bar{1}\,\bar{1}] \\ [1\bar{1}0]\,[\bar{1}\,10] \\ [10\bar{1}]\,[\bar{1}\,01] \\ [01\,\bar{1}]\,[0\bar{1}\,1] \end{cases}$$

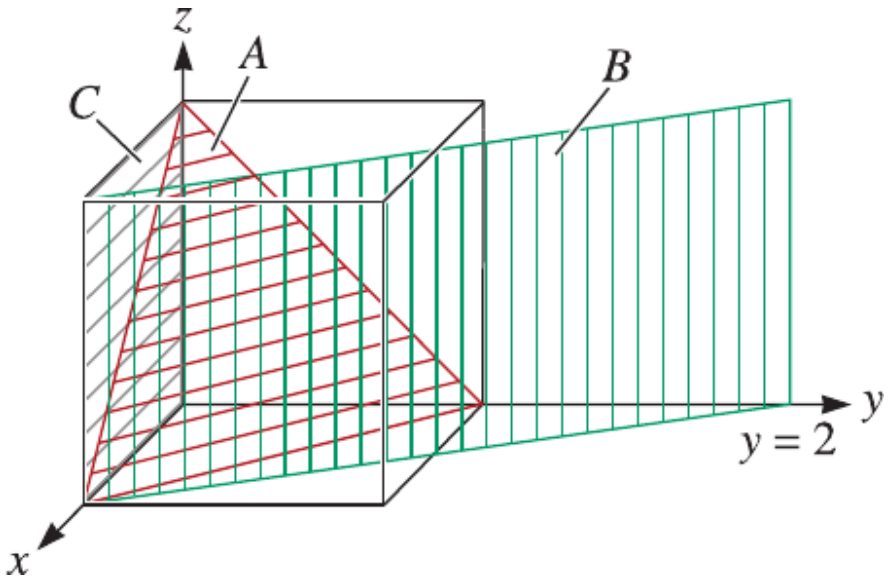


Miller indices are used to identify planes, as described in the following procedure:

1. Identify the points at which the plane intercepts the x, y, and z coordinates in units of lattice parameters. If the plane passes through the origin, the origin of the coordinate system must be moved to a nearby unit cell.
2. Take reciprocals of these intercepts.
3. Clear fractions but do not reduce to lowest integers.
4. Enclose the resulting numbers in parentheses ( ). Negative numbers should be written with a bar over the number.



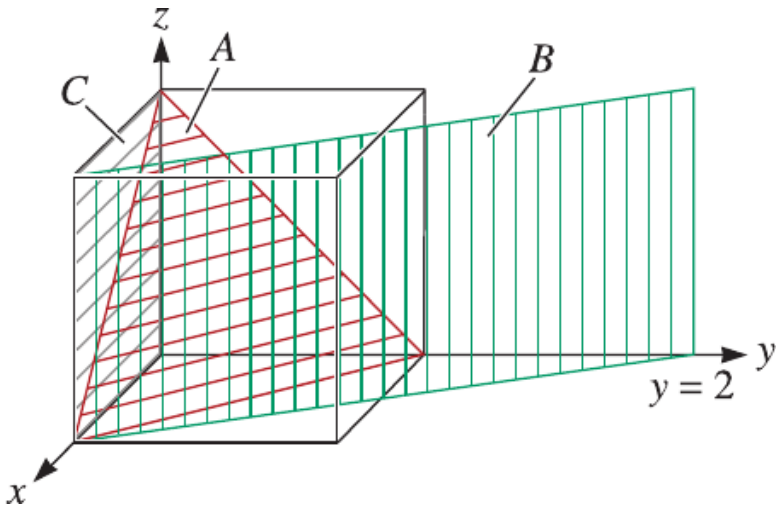
Determine the Miller indices of planes A, B, and C.



Solution: Plane A

1. $x = 1,\ y = 1,\ z = 1$
2. $\frac{1}{x} = 1,\ \frac{1}{y} = 1,\ \frac{1}{z} = 1$
3. No fractions to clear
4. $(1\,1\,1)$

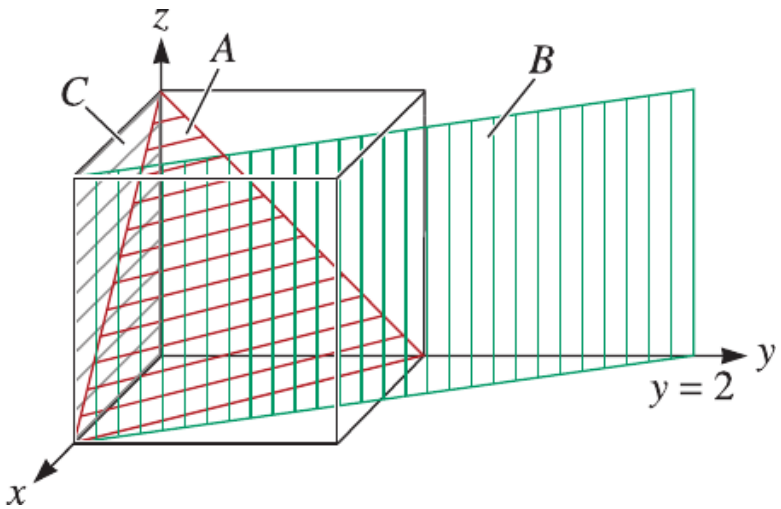
Determine the Miller indices of planes A, B, and C.



Solution: Plane B

1. The plane never intercepts the z axis. Therefore $x = 1$, $y = 2$, $z = \infty$
2. $\frac{1}{x} = 1$, $\frac{1}{y} = \frac{1}{2}$, $\frac{1}{z} = 0$
3. Clear fractions by multiplying by 2: $\frac{1}{x} = 2$, $\frac{1}{y} = 1$, $\frac{1}{z} = 0$
4. $(2\,1\,0)$

Determine the Miller indices of planes A, B, and C.



Solution: Plane C

1. The origin must be moved, since the plane passes through 0, 0, 0. By moving the origin one lattice parameter in the y-direction: $x = \infty$, $y = -1$, $z = \infty$
2. $\frac{1}{x} = 0$, $\frac{1}{y} = -1$, $\frac{1}{z} = 0$
3. No fraction to clear.
4. $(0\,\bar{1}\,0)$

- Planes and their negatives are identical (not the case for directions) because they are parallel. Hence, $(0\,2\,0) = (0\,\bar{2}\,0)$.
- Planes and their multiples are not identical (the opposite for directions).
- In each unit cell, planes of a family represent groups of equivalent planes that have their particular indices because of the orientation of the coordinates.
- These groups of similar planes are represented with { }.

Planes of the form {110} in cubic systems

$$\{110\}\begin{cases}(110)\\(101)\\(011)\\(1\bar{1}0)\\(10\bar{1})\\(01\bar{1})\end{cases}$$

Note: The negatives of the planes are not unique planes.

At the end of the section, you should be able to:

- Calculate the linear atomic density on a direction for cubic crystals.
- Calculate the planar atomic density on a given crystal plane for cubic crystals.



Linear atomic density

- Linear density (ρ_L) is defined as the number of atoms per unit length in a given crystallographic direction.
- Only the atoms with their centres lie in the crystallographic direction are counted.

$$\rho_L = \frac{N}{l}$$

where N is the number of atoms with their centers lie in the direction and l is the length of the line segment.

- The direction with a maximum atomic packing density is known as the **closed-packed direction**.



Linear atomic density

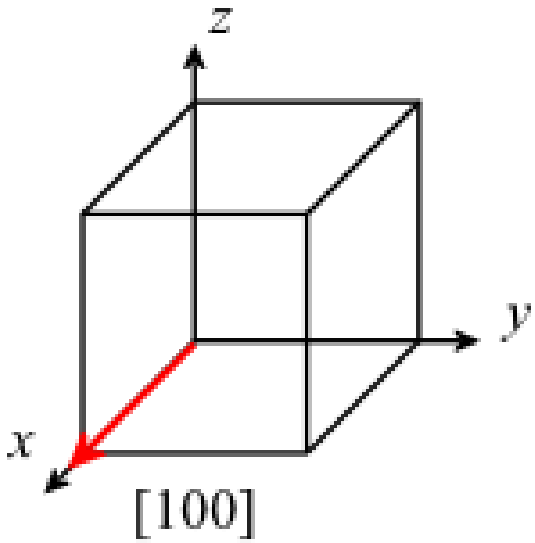
Problem

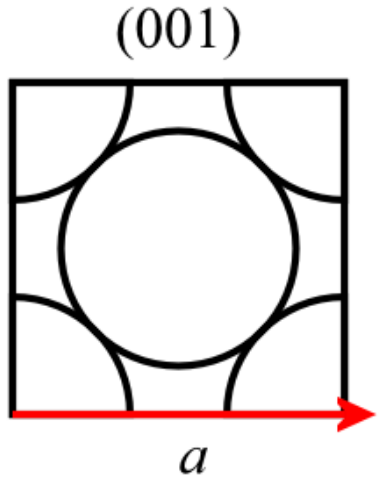
Exercise

1. Determine the linear density for FCC nickel which has a lattice parameter of $0.351\,67$ nm in the [100], [110], and [111] directions.
2. Which of these directions is close packed?

Linear atomic density

Linear density along FCC [100]





$$N = \left(\frac{1}{2} + \frac{1}{2}\right) = 1$$

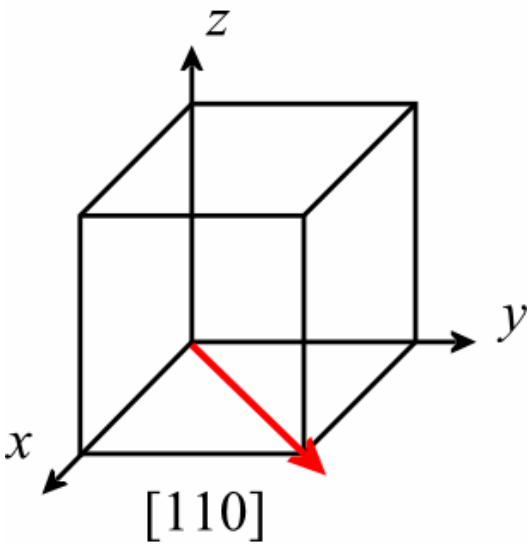
$$l = a = 0.351\,67\,\text{nm}$$

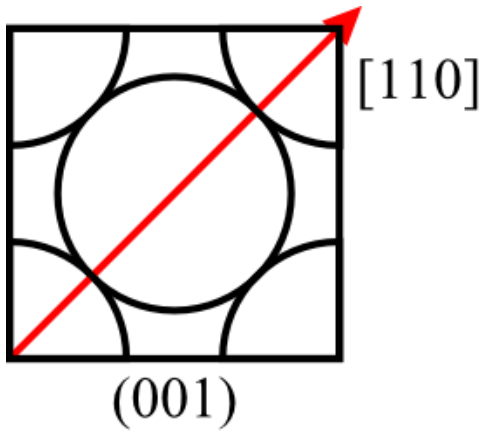
$$\rho_L = \frac{N}{l} = \frac{1}{0.35167}$$

$$= 2.8436\,\text{atoms/nm}$$

Linear atomic density

Linear density along FCC [110]





$$N = \left(\frac{1}{2} + 1 + \frac{1}{2}\right) = 2 \text{ atoms}$$

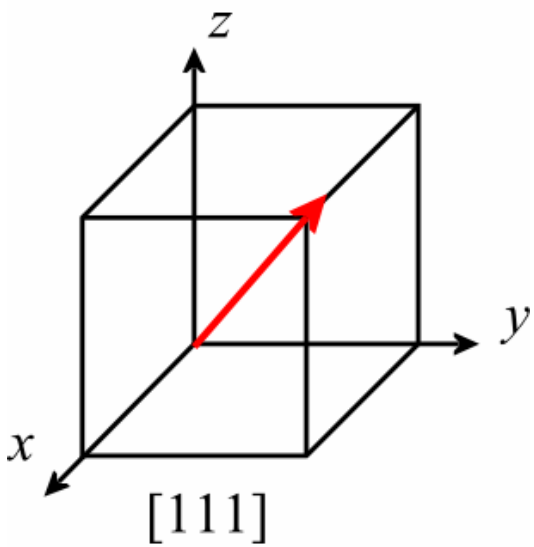
$$l = \sqrt{2}a = \sqrt{2} \times 0.351\,67 \text{ nm}$$

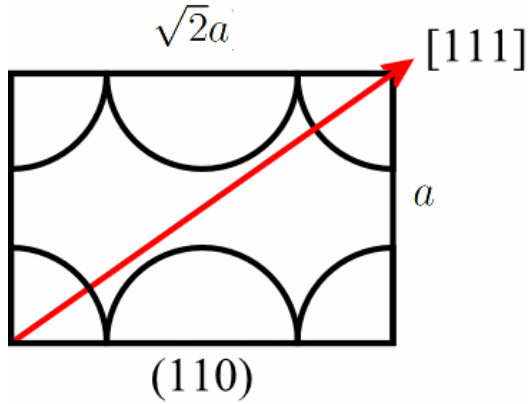
$$\rho_L = \frac{N}{l} = \frac{2}{\sqrt{2} \times 0.35167} \text{atoms/nm}$$

$$= 4.0214 \text{ atoms/nm}$$

Linear atomic density

Linear density along FCC [111]





$$N = \frac{1}{2} + \frac{1}{2} = 1 \text{ atoms}$$

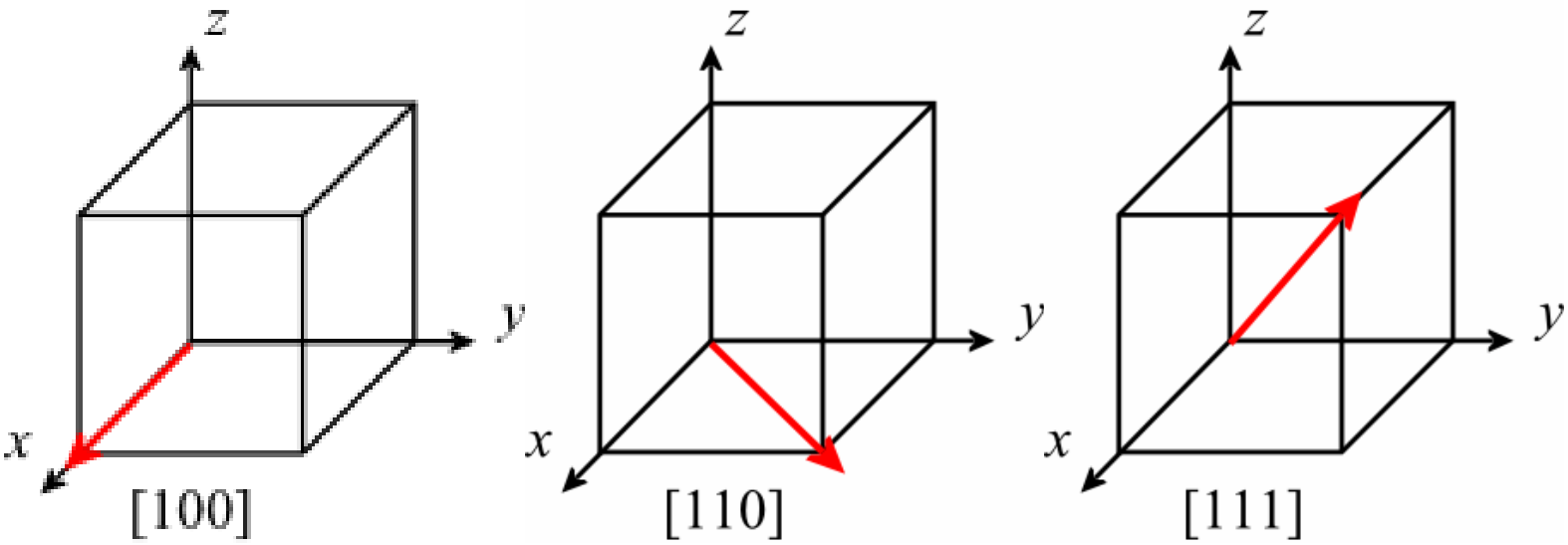
$$l = \sqrt{3}a = \sqrt{3} \times 0.351\,67 \text{ nm}$$

$$\rho_L = \frac{N}{l} = \frac{1}{\sqrt{3} \times 0.35167} \text{atoms/nm}$$

$$= 1.6417 \text{ atoms/nm}$$

Linear atomic density

Close pack direction



- The highest linear density is in the [110] direction (4.0214 atoms/nm).
- The [110] is the closest packed direction of the three directions considered in the problem.
- Because atoms touch each other in the [110] direction, it will be the highest possible linear density for the Ni FCC crystal!
- In fact, [110] direction will be the close packed direction for any FCC crystal.

Planar atomic density

- Planar density (ρ_P) is defined as the number of atoms per unit area on a crystallographic plane.
- Only the atoms with their centres lie on the crystallographic plane are counted.

$$\rho_P = \frac{N}{A}$$

where N is the number of atoms with their centers on the plane and A is the area of the plane.

- The area with a maximum atomic packing density is known as a **closed-packed plane**.



Planar atomic density

Problem

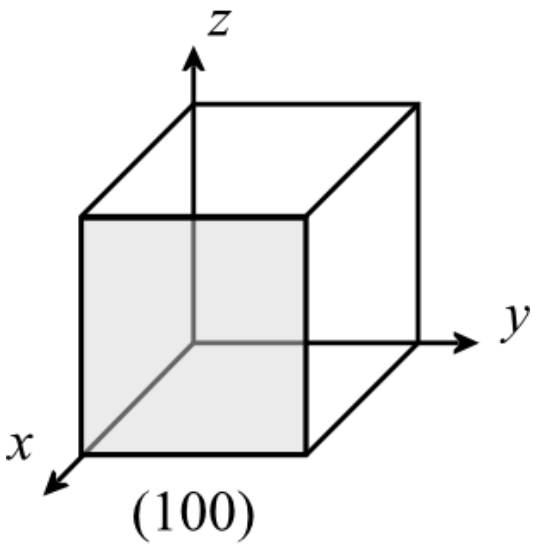
Exercise

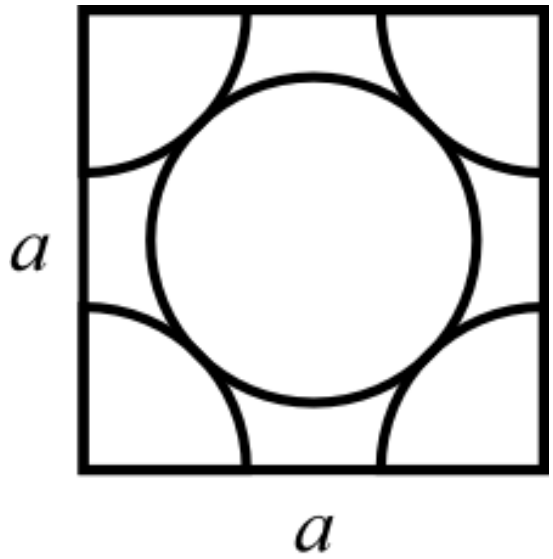
1. Determine the planar density for FCC nickel which has a lattice parameter of $0.351\,67\,\text{nm}$ in the (100), (110), and (111) planes.
2. Which of these planes is close packed?



Planar atomic density

Planar density on FCC (100)





$$N = \left(\frac{1}{4} \times 4\right) + 1 = 2$$

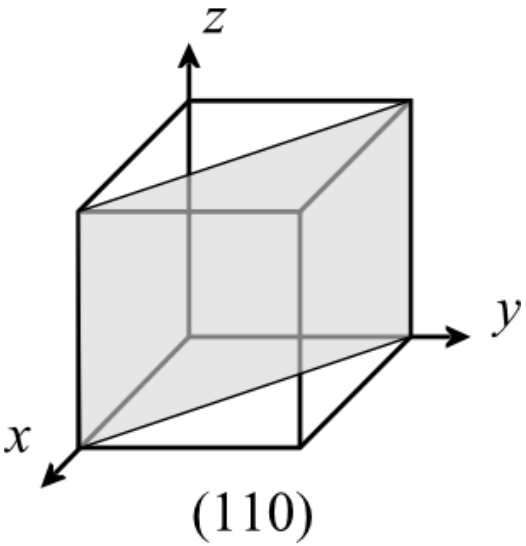
$$A = a^2$$

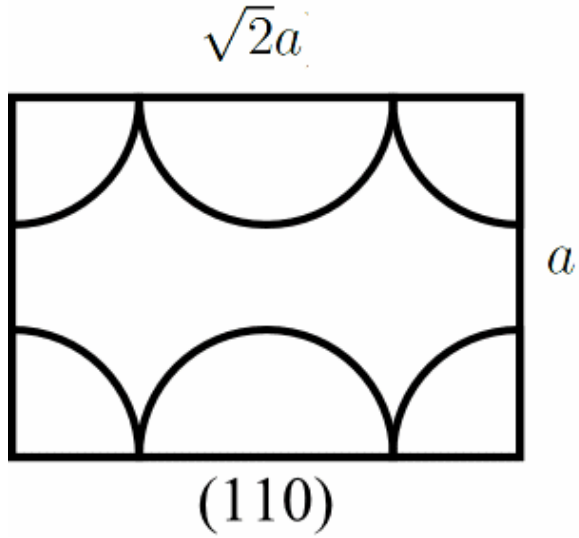
$$\rho_A = \frac{N}{A} = \frac{N}{a^2} = \frac{2}{(0.35167)^2}$$

$$= 16.1718 \,\text{atoms/nm}^2$$

Planar atomic density

Planar density on FCC (110)





$$N = \left(\frac{1}{4} \times 4\right) + \left(\frac{1}{2} \times 2\right) = 2$$

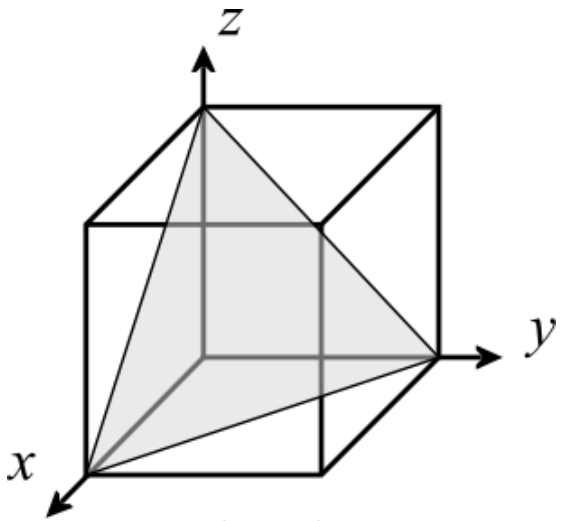
$$A = a \times \sqrt{2}a = \sqrt{2}a^2$$

$$\rho_A = \frac{N}{A} = \frac{2}{\sqrt{2}a^2} = \frac{2}{\sqrt{2} \times (0.35167)^2}$$

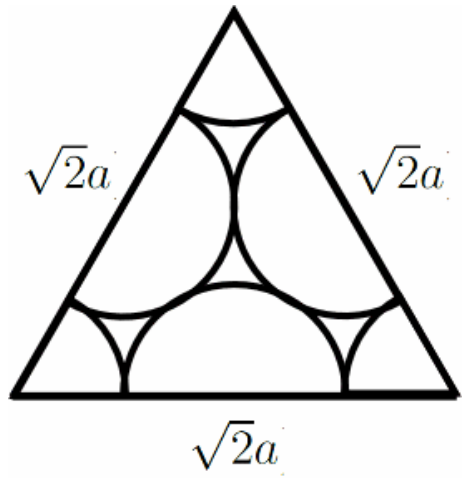
$$= 11.4352 \,\text{atoms/nm}^2$$

Planar atomic density

Planar density on FCC (111)



(111)



(111)

$$N = \left(\frac{1}{6} \times 3\right) + \left(\frac{1}{2} \times 3\right) = 2$$

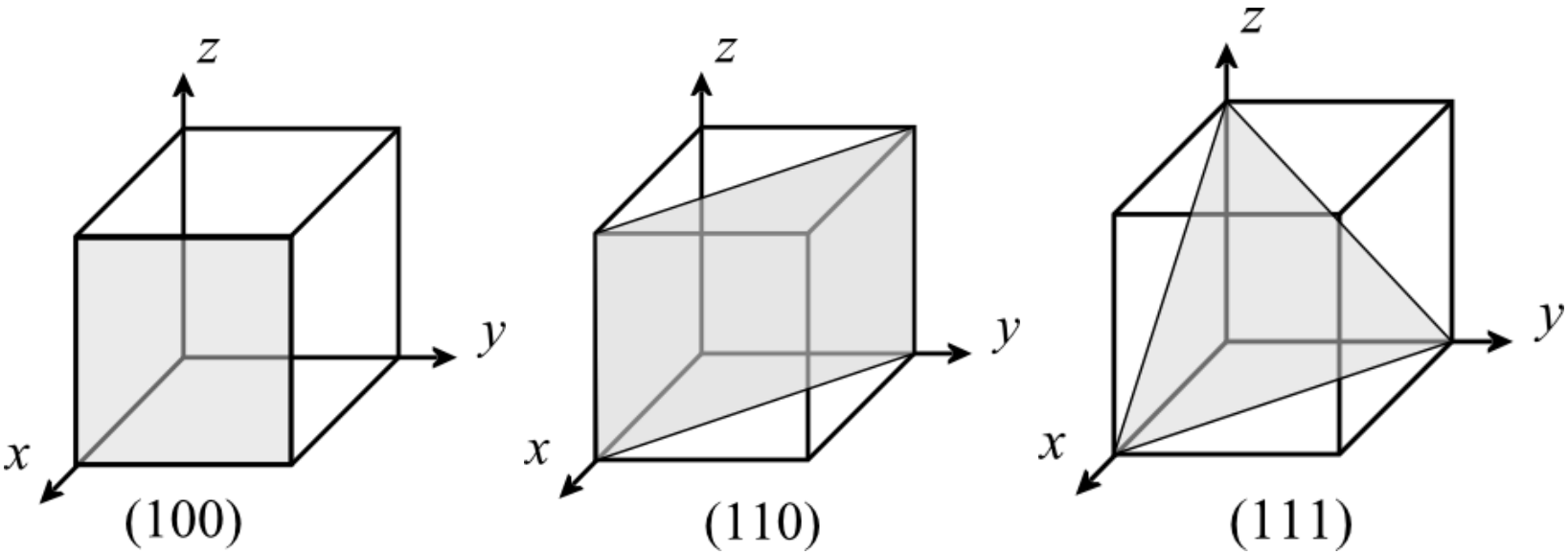
$$A = \frac{1}{2} \times \sqrt{2}a \times (\sqrt{2}a \sin 60^\circ) = \frac{\sqrt{3}}{2}a^2$$

$$\rho_A = \frac{N}{A} = \frac{2}{(\sqrt{3}/2)a^2} = \frac{4}{\sqrt{3} \times (0.35167)^2}$$

$$= 18.6736 \text{ atoms/nm}^2$$

Planar atomic density

Planar density on FCC (111)



- The highest planar density is in the (111) plane ($18.6736\,\text{atoms/nm}^2$).
- The (111) is the closest packed plane of the three planes considered in the problem.
- Because atoms touch each other in the (111) plane, it will be the highest possible planar density for the Ni FCC crystal!
- In fact, (111) plane will be the closest packed plane of any FCC crystal.